

GENERAL PROPERTIES OF MANY PARTICLE SYSTEMS AT LOW TEMPERATURES

NISHTHA TIKALAL

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1. QUASI PARTICLES

Definition 1.1. Statistical physics studies the behaviour of a system containing a large # of particles.

When we consider things macroscopically, it often arises from microscopic interactions made by the particles in the system.

Overall macroscopic properties are determined by certain averages.

e.g.: we understand a lot of thermodynamics & thermodynamic systems using statistical physics.

In thermodynamics, the 3 main properties that make up a macroscopic system are:

P = pressure

T = temperature

N = # of particles (average)

In quantum mechanics, we characterize such systems by E_n , the energy of a system.

Let us define a system (closed):

In itself, the subsystem is very large, but the forces interacting between the individual particles are very small. $\therefore$ we can conclude that the subsystem too, is closed. It is also characterised by its own set of energy values. Note that not only do the particles in the subsystem interact amongst themselves, but also interact with the particles from the entire closed system itself. Thus, we cannot say that the subsystem has a fixed energy state & a fixed # of particles.

Microscopic derivations are based off Gibbs distribution.

Definition 1.2. Gibbs Distribution: Gives the probability of finding the subsystem in the energy state E_{nN} , where n is the # of particles in the subsystem & N is the # of particles in the closed system.

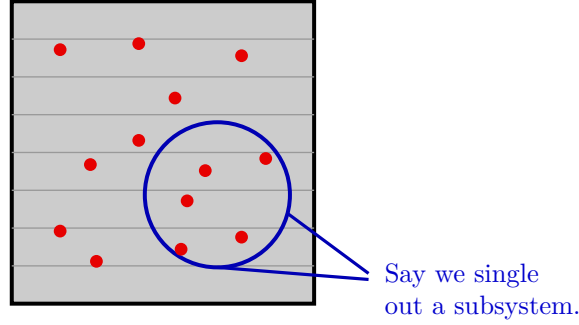


Figure 1: Schematic representation of a system and a singled-out subsystem.

$$w_{nN} = Z^{-1} e^{-(E_{nN} - \mu N)/T} \quad \dots (1)$$

Here, T is the absolute temperature, μ is the chemical potential, & Z is a normalisation condition determined by the condition:

$$\sum_{n,N} w_{nN} = 1 \quad \dots (2)$$

By substitution of (2) into (1),

$$\begin{aligned} \sum_{n,N} w_{nN} &= \sum_{n,N} Z^{-1} e^{-(E_{nN} - \mu N)/T} \\ 1 &= \sum_{n,N} Z^{-1} e^{-(E_{nN} - \mu N)/T} \\ 1 &= \sum_{n,N} \frac{1}{Z} e^{-(E_{nN} - \mu N)/T} \\ Z &= \sum_{n,N} e^{-(E_{nN} - \mu N)/T} \quad \dots (3) \end{aligned}$$

Z is known as the grand partition function. This immediately elucidates other thermodynamic functions by the formula:

$$\Omega = -T \ln(Z) \quad \dots, (4)$$

where Ω is the potential involving μ , V , & T .

These formulas are most commonly applied to the case of ideal gases since (1) is simply just a sum of energies from the individual particles. Doing this calculation over & over again may seem extremely exhaustive & unrealistic. This is only usually applied when:

- (i) Interactions are weak.
- (ii) Perturbation stops at either first or second order approximations.

As $T \rightarrow 0$, the important energy levels in the partition function are the weakly excited states, whose values are comparable to the ground state energies.

Recall by (1),

$$\begin{aligned}
w_{nN} &= Z^{-1} e^{-(E_{nN} - \mu N)/T} \\
w_{nN} e^{(E_{nN} - \mu N)/T} &= Z^{-1} \\
e^{(E_{nN} - \mu N)/T} &= Z^{-1} \cdot w_{nN}^{-1} \\
\ln \left(e^{(E_{nN} - \mu N)/T} \right) &= \ln \left(Z^{-1} w_{nN}^{-1} \right) \\
\frac{E_{nN} - \mu N}{T} &= \ln \left(Z^{-1} w_{nN}^{-1} \right) \\
\frac{E_{nN} - \mu N}{T} &= \ln \left(Z^{-1} \right) + \ln \left(w_{nN}^{-1} \right) \\
\frac{E_{nN} - \mu N}{T} &= -1 \ln(Z) + -1 \ln(w_{nN}) \\
\frac{E_{nN} - \mu N}{T} &= -1 (\ln(Z) + \ln(w_{nN})) \\
T &= \frac{E_{nN} - \mu N}{-\ln(Z) - \ln(w_{nN})}
\end{aligned}$$

As $T \rightarrow 0$, the system completely settles into the ground state ($n = 0$). The probability of being in the ground state becomes certain ($w_0 \rightarrow 1$).

Because $\ln(1) = 0$, this leaves us with:

$$T = \frac{E_{nN} - \mu N}{-\ln(Z)}$$

Notice that by re-arrangement,

$$\begin{aligned}
T[-\ln(Z)] &= E_{nN} - \mu N \\
-T \ln(Z) &= E_{nN} - \mu N \\
\Omega &= E_{nN} - \mu N
\end{aligned}$$

In this region the character of the energy spectrum can be ascertained to some detail.

To study this, we look at the excitation of lattice vibrations in a crystal. As long as the vibrations are small, then we view the lattice as a set of coupled harmonic oscillators.

Recall that from quantum mechanics, the energy of a harmonic oscillator is:

$$E = \sum_{i=1}^{3N} \omega_i \left(n_i + \frac{1}{2} \right)$$

The vibrations within the lattice can be described as monochromatic plane waves.

Definition 1.3. Monochromatic plane waves: ideal, continuous waves that travel in one direction & oscillates at exactly one frequency.

With these monochromatic waves, there is an associated wave vector (k) & frequency (ω_s), where the index of s denotes the type of wave. Since there is the possibility of more than one kind of wave propagating in the lattice crystal, the frequency is not a single-valued function. Rather it is a function of several wave-vectors:

$$f = \omega_s(k)$$

For small momenta, branches are characterised by linear dependency of k :

$$\begin{aligned} f &= \omega_s(k) \\ \omega_s(k) &= u_s(\theta, \varphi)|k| \end{aligned}$$

While we can use this model to determine thermodynamic & kinetic characteristics of the lattice, there is more convenience in picking a different model. This model came forth in the advent of quantum mechanics.

- i) First Quantisation: particles behave like waves
- ii) Second Quantisation: waves behave like particles

Per the second quantization, let every plane wave correspond to a set of moving particles with momentum determined by k & energy by $\omega_s(k)$. Therefore excited states in a lattice are simply an aggregate of particles moving within the volume of the crystal.

$\hookrightarrow$ aggregate of particles = phonons.

For now, we note that this model also allows linear dependency on k , which leads us to a number of qualitative conclusions about the lattice. Quantitatively, we use the isotropic Debye model. In this model, we don't assume a number of branches. Rather, we have an isotropic body that consists of the following:

- i) Longitudinal phonons with energy $\omega_l(k) = u_l k$
- ii) Transverse phonons of two possible polarisations & dependence $\omega_t(k) = u_t k$

Additionally, we assume that the momenta of the phonons don't exceed an upper bound k_D .

$$k_D \sim \frac{1}{a} \quad \} \text{ interatomic distance}$$

It is important to note that this phonon model is an approximation in low-temperature regimes. As T increases, so does the interaction between phonons.

From the discussion on the lattice, we turn to the general case. In order to study a model of an energy spectrum, we make the assumption that the structure of the energy levels are that of an ideal gas. We know that the energy level is obtained by the sum of energies from a collection of quasi-particles or elementary excitations.

Definition 1.4. Elementary excitations: result of collective interactions of the particles of a system, & therefore pertain to the system as a whole; not as separate particles.

$$\text{elementary excitations} \neq \# \text{ of particles}$$

Energy can be divided into two types:

Elementary excitations do not correspond to statistical states, but instead represent wave packets.

Definition 1.5. Wave Packets: superpositions of large numbers of exact stationary states with a narrow spread in energy.

Spread in the packet / attenuation / damping is a result of interaction between quasi-particles.

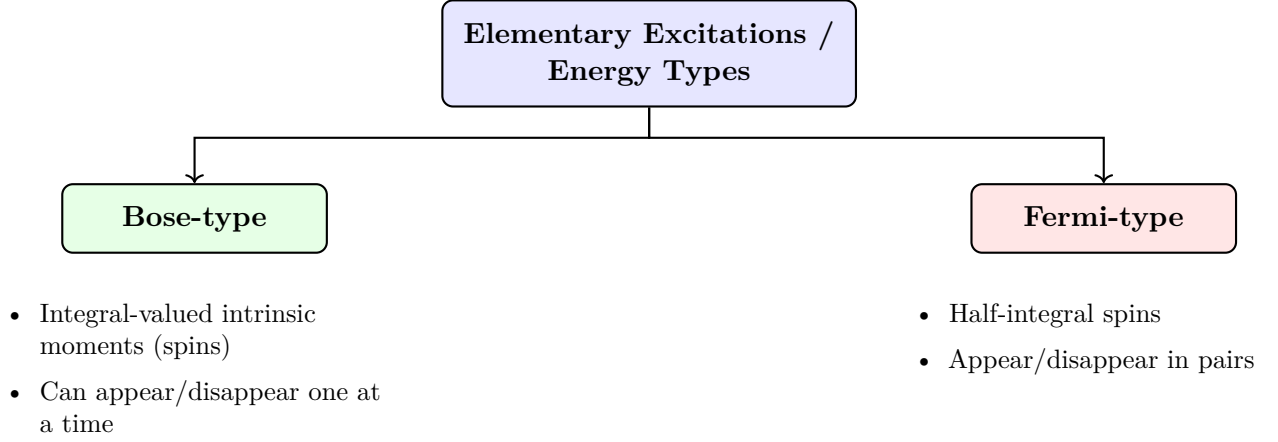


Figure 2: Classification of excitation energy types.

2. THE SPECTRUM OF A BOSE LIQUID

Definition 2.1. Bose liquid: A liquid consisting of atoms with integral-valued spins.

The dependence of the excitation energies of a Bose liquid on the momentum (small-valued) can be determined by general considerations. Small values of p corresponds to long-wavelength oscillations of the liquid (which are sound waves).

$\therefore$ for small p , elementary excitations = acoustic quanta (phonons). Note that the acoustic frequency ω is related to the wave vector by:

$$\omega = uk \quad (5)$$

where u is the velocity of sound and k is the wave vector. Using (5), we can determine the excitation energy as a function of momentum.

Recall:

$$\begin{aligned} E &= \hbar\omega \\ p &= \hbar k \rightarrow \hbar = \frac{p}{k} \end{aligned}$$

By substitution of $\hbar$ & ω into E , we obtain:

$$\begin{aligned} E &= \frac{p}{k} \cdot uk \\ E &= up \end{aligned}$$

This yields the final relation:

$$\varepsilon(p) = up \quad (6)$$

We see that the energy is linearly dependent on momentum. However, as $\varepsilon(p)$ increases, linear dependence on p ceases. We shift our attention to the energy of a liquid as a function of its density $\rho(r)$ & its hydrodynamic velocity $\mathbf{v}(r)$:

$$E(\rho, \mathbf{v}) = \frac{1}{2} \int \rho v^2 dr + E^{(1)}(\rho) \quad (7)$$

where $E^{(1)}(\rho)$ is the part of energy that is velocity-independent.

Concerning small oscillations,

$$\rho(r) = \bar{\rho} + \delta\rho(r) \quad (8)$$

Here, $\bar{\rho}$ is the equilibrium density (the "average") and $\delta\rho(r)$ is a small quantity describing the oscillations.

By definition,

$$\bar{\rho} = \frac{1}{V} \int \rho(r) \, dr \quad (9)$$

$$\int \delta\rho(r) \, dr = 0 \quad (10)$$

Note that V is the volume of the liquid. Within quantities of the second order, we can re-write $E^{(1)}(\rho)$ from (7) using $\bar{\rho}$.

Recall (7):

$$\frac{1}{2} \int \rho v^2 \, dr + E^{(1)}(\rho)$$

Substitution of (8) yields:

$$\underbrace{\frac{1}{2} \int (\bar{\rho} + \delta\rho(r)) v^2 \, dr}_{I_A} + \underbrace{E^{(1)}(\bar{\rho} + \delta\rho(r))}_{\text{approximated by an expression; for now, this is } I_B}$$

Evaluating I_A ,

$$\begin{aligned} & \frac{1}{2} \int [\bar{\rho} v^2 + \delta\rho(r) v^2] \, dr \\ &= \frac{1}{2} \int \bar{\rho} v^2 \, dr + \frac{1}{2} \int \delta\rho(r) v^2 \, dr \end{aligned}$$

By (10),

$$= \frac{1}{2} \int \bar{\rho} v^2 \, dr$$

Looking at I_B , we approximate $E^{(1)}(\rho)$ as the following, which is uniquely determined by the properties of the liquid (homogeneous, isotropic):

$$E^{(1)}(\rho) = E^{(1)}(\bar{\rho}) + \int \psi(r) \delta\rho(r) \, dr + \frac{1}{2} \iint \varphi(r, r') \delta\rho(r) \delta\rho(r') \, dr \, dr' \quad (11)$$

Under these conditions,

$$\begin{aligned} \psi(r) &= \text{constant} = \psi \\ \varphi(r, r') &= \varphi(|r - r'|) \end{aligned}$$

$\therefore$

$$E^{(1)}(\rho) = E^{(1)}(\bar{\rho}) + \underbrace{\int \delta\rho(r) \, dr}_0 + \frac{1}{2} \iint \varphi(|r - r'|) \delta\rho(r) \delta\rho(r') \, dr \, dr'$$

Putting I_A and I_B together, (7) becomes:

$$\begin{aligned} E(\rho, \mathbf{v}) &= \frac{1}{2} \int \rho v^2 \, d\mathbf{r} + E^{(1)}(\rho) \\ &= \frac{1}{2} \bar{\rho} \int v^2 \, d\mathbf{r} + \frac{1}{2} \iint \varphi(|r - r'|) \delta\rho(r) \delta\rho(r') \, d\mathbf{r} \, d\mathbf{r}' + E^{(1)}(\bar{\rho}) \end{aligned} \quad (12)$$

Velocity $\mathbf{v}$ is related to density oscillations by the continuity equation:

$$\dot{\rho} + \operatorname{div}(\rho \mathbf{v}) = 0$$

$$\dot{\rho} + \bar{\rho} \operatorname{div}(\mathbf{v}) = 0 \quad (13)$$

We let the following be described by the Fourier components:

$$\begin{cases} \delta\rho(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{p}} \rho_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{r}} \\ \mathbf{v}(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{p}} \mathbf{v}_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{r}} \\ \varphi(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{p}} \varphi_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{r}} \end{cases} \quad (14)$$

We also note that small oscillations of the liquid are always longitudinal. This means the velocity of a wave of a wave vector $\mathbf{p}$ is always directed along $\mathbf{p}$:

$$\mathbf{v}_{\mathbf{p}} = a_{\mathbf{p}} \mathbf{p} \quad (15)$$

It follows that:

Recall (13): $\dot{\rho} + \bar{\rho} \operatorname{div}(\mathbf{v})$. Using (14),

$$\begin{aligned} \mathbf{v}(\mathbf{r}) &= \frac{1}{V} \sum_{\mathbf{p}} \mathbf{v}_{\mathbf{p}} e^{i\mathbf{p} \cdot \mathbf{r}} \\ \operatorname{div}[\mathbf{v}(\mathbf{r})] &= \nabla \cdot \mathbf{v}(\mathbf{r}) \\ &= \frac{\partial[\mathbf{v}(\mathbf{r})]}{\partial \mathbf{r}} \\ &= i\mathbf{p} e^{i\mathbf{p} \cdot \mathbf{r}} \\ &= \frac{1}{V} \sum_{\mathbf{p}} \mathbf{v}_{\mathbf{p}} \cdot i\mathbf{p} e^{i\mathbf{p} \cdot \mathbf{r}} \end{aligned}$$

By the expression above, (13) becomes:

$$\dot{\rho} + \bar{\rho} \left(\frac{1}{V} \sum_{\mathbf{p}} \mathbf{v}_{\mathbf{p}} \cdot i\mathbf{p} e^{i\mathbf{p} \cdot \mathbf{r}} \right)$$

Also recall from (8),

$$\rho(\mathbf{r}) = \bar{\rho} + \delta\rho(\mathbf{r})$$

Taking the time derivative,

$$\begin{aligned} \dot{\rho}(\mathbf{r}) = \dot{\rho}(r) &= \frac{\partial}{\partial t} (\bar{\rho} + \delta\rho(\mathbf{r})) \\ &= \frac{\partial \bar{\rho}}{\partial t} + \frac{\partial[\delta\rho(\mathbf{r})]}{\partial t} \end{aligned}$$

We note $\bar{\rho}$ is the equilibrium density (coordinate-independent). Therefore, taking its time derivative $\equiv 0$.

For the second component, $\frac{\partial[\delta\rho(\mathbf{r})]}{\partial t}$, we take the derivative of the expression from (14). Recall:

$$\begin{aligned}\delta\rho(\mathbf{r}) &= \frac{1}{V} \sum_{\mathbf{p}} \rho_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}} \\ \frac{\partial[\delta\rho(\mathbf{r})]}{\partial t} &= \frac{1}{V} \sum_{\mathbf{p}} \dot{\rho}_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}}\end{aligned}$$

Substituting the solved expressions of $\dot{\rho}$ & $\text{div}(\mathbf{v})$ into (13), we obtain:

$$\begin{aligned}\dot{\rho} + \bar{\rho}(\text{div } \mathbf{v}) &= 0 \\ \frac{1}{V} \sum_{\mathbf{p}} \dot{\rho}_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}} + \bar{\rho} \left(\frac{1}{V} \sum_{\mathbf{p}} i(\mathbf{p} \cdot \mathbf{v}_{\mathbf{p}}) e^{i\mathbf{p}\cdot\mathbf{r}} \right) &= 0\end{aligned}$$

By factorization,

$$\frac{1}{V} \sum_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{r}} (\dot{\rho}_{\mathbf{p}} + \bar{\rho}(i\mathbf{p} \cdot \mathbf{v}_{\mathbf{p}})) = 0$$

To address the dot product in parentheses, we substitute (15):

$$\bar{\rho}(\mathbf{p} \cdot \mathbf{v}_{\mathbf{p}}) = \bar{\rho}(i\mathbf{p} \cdot a_{\mathbf{p}}\mathbf{p}) = \bar{\rho}(ia_{\mathbf{p}}p^2)$$

$\therefore$ The full expression is:

$$\begin{aligned}\dot{\rho}_{\mathbf{p}} + \bar{\rho}(ia_{\mathbf{p}}p^2) &= 0 \\ -\dot{\rho}_{\mathbf{p}} &= \bar{\rho}ip^2a_{\mathbf{p}} \\ a_{\mathbf{p}} &= \frac{-\dot{\rho}_{\mathbf{p}}}{i\bar{\rho}p^2}\end{aligned}\tag{16}$$

Substitution into (15) yields another expression for $\mathbf{v}_{\mathbf{p}}$:

$$\begin{aligned}\mathbf{v}_{\mathbf{p}} &= a_{\mathbf{p}}\mathbf{p} \\ &= \frac{-\dot{\rho}_{\mathbf{p}}}{i\bar{\rho}(p^2)}\mathbf{p}\end{aligned}$$

$\therefore$

$$\mathbf{v}_{\mathbf{p}} = i \frac{\dot{\rho}_{\mathbf{p}}}{\bar{\rho}} \cdot \frac{\mathbf{p}}{p^2}\tag{17}$$

By (17), eq. (12) becomes:

$$E = \underbrace{E^{(1)}(\bar{\rho})}_{\text{energy of an unperturbed liquid}} + \underbrace{\frac{1}{V} \sum_{\mathbf{p}} \left(\frac{|\dot{\rho}_{\mathbf{p}}|^2}{2\bar{\rho}p^2} + \frac{1}{2}\varphi_{\mathbf{p}}|\rho_{\mathbf{p}}|^2 \right)}_{\text{energy of a harmonic oscillator, where frequency:}}\tag{18}$$

$$\omega_{\mathbf{p}}^2 = \bar{\rho}p^2\varphi_{\mathbf{p}}\tag{19}$$

(18) tells us that all small oscillations of the liquid decompose into elementary excitations, to which we describe by harmonic oscillators. Recall that from quantum mechanics, the energy of the quantum mechanic oscillator is given by:

$$\varepsilon_{\mathbf{p}} = \omega_{\mathbf{p}} \left(n + \frac{1}{2} \right) \quad ; \quad (n = 0, 1, 2, \dots)$$

We can see that the equation above & eq. (18) are structurally identical in modelling elementary excitation. The dependence of $\varepsilon(p)$ on the momentum, if recalled, is given by (19) as well as:

$$\varepsilon(p) = \omega_{\mathbf{p}} \tag{20}$$

In order to make (18) more complete, we take into account the zero-point energy of the oscillators, which we know from:

$$\begin{aligned} \varepsilon(p) &= \omega_{\mathbf{p}} \left(n + \frac{1}{2} \right) \\ &= \frac{1}{2} \omega_{\mathbf{p}} \end{aligned}$$

$\therefore$ The ground state energy of the Bose liquid becomes:

$$E_0 = E^{(1)}(\bar{\rho}) + \sum_{\mathbf{p}} \frac{1}{2} \omega_{\mathbf{p}}$$

We see that:

$$\frac{1}{V} \sum_{\mathbf{p}} \left(\frac{|\rho_{\mathbf{p}}|^2}{2\bar{\rho}p^2} + \frac{1}{2} \varphi_{\mathbf{p}} |\rho_{\mathbf{p}}|^2 \right) = \frac{\omega_{\mathbf{p}}}{2}$$

so,

$$\frac{V \cdot \omega_{\mathbf{p}}}{2} = \sum_{\mathbf{p}} \frac{|\rho_{\mathbf{p}}|^2}{2\bar{\rho}p^2} + \frac{1}{2} \varphi_{\mathbf{p}} |\rho_{\mathbf{p}}|^2$$

By the LHS of the above equation, we apply virial theorem! Recall that by virial theorem,

$$T = U$$

Thus,

$$\begin{aligned} \frac{V\omega_{\mathbf{p}}}{2} &= \frac{1}{2} \rho_{\mathbf{p}} |\mathbf{v}_{\mathbf{p}}|^2 + \frac{1}{2} \varphi_{\mathbf{p}} |\rho_{\mathbf{p}}|^2 \\ &= \varphi_{\mathbf{p}} |\rho_{\mathbf{p}}|^2 \end{aligned} \tag{21}$$

From eq. (21),

$$\frac{V\omega_{\mathbf{p}}}{2} = \varphi_{\mathbf{p}} |\rho_{\mathbf{p}}|^2$$

$$\varphi_{\mathbf{p}} = \frac{V\omega_{\mathbf{p}}}{2|\rho_{\mathbf{p}}|^2}$$

$$\omega_{\mathbf{p}}^2 = \bar{\rho} p^2 \varphi_{\mathbf{p}}$$

$$= \bar{\rho} p^2 \cdot \frac{V\omega_{\mathbf{p}}}{2|\rho_{\mathbf{p}}|^2}$$

$$\omega_{\mathbf{p}} = \frac{\bar{\rho} p^2 \cdot V}{2|\rho_{\mathbf{p}}|^2}$$

By (20),

$$\begin{aligned}\varepsilon(p) &= \omega_{\mathbf{p}} \\ &= \frac{p^2}{2mS(p)}\end{aligned}\tag{22}$$

where $S(p)$ is defined as:

$$S(p) = \frac{|\rho_{\mathbf{p}}|^2}{Vm\bar{\rho}}\tag{23}$$

↓ density correlation function

$$S(\mathbf{r} - \mathbf{r}') = \frac{[n(\mathbf{r}) - \bar{n}][n(\mathbf{r}') - \bar{n}]}{\bar{n}} \quad ; \quad n(\mathbf{r}) = \frac{1}{m}\rho(\mathbf{r})\tag{24}$$

Note n is the number of particles. Recall that for small momenta, the excitation energy was a linear function of momentum (i.e., $\varepsilon(p) = up$). This implies (by (22)), that $S(p)$ also depends linearly on momentum:

$$S \approx \frac{p}{2mu}\tag{25}$$

We obtain this by:

Recall (22):

$$\varepsilon(p) = \omega_{\mathbf{p}} = \frac{p^2}{2mS(p)}$$

Also recall that: $\varepsilon(p) = up$.

∴ equating the two expressions yield:

$$up = \frac{p^2}{2mS(p)}$$

By a slight variable re-arrangement,

$$S \approx \frac{p}{2mu}, \quad \text{which was noted in (24).}$$

In regions of large momenta (or small distances), the function $S(\mathbf{r})$ has the form:

$$S(\mathbf{r}) = \delta(\mathbf{r}) + \nu(\mathbf{r})\tag{26}$$

Note that ν has no singularities as $\mathbf{r} \rightarrow 0$. Thus, in Fourier components,

$$S(\mathbf{p}) = 1 + \nu(\mathbf{p}),\tag{27}$$

where $\nu(\mathbf{p}) \rightarrow 0$ as $p \rightarrow \infty$. We can gather that by (26) $S(p) \approx 1$ & (22) becomes:

$$\varepsilon(p) \approx \frac{p^2}{2m} \quad // \text{ identical to the energy of a free atom!}$$

Again, we note that these formulae are only valid for small momenta. In regions of low T , the most important part of the energy spectrum corresponds to small p (or phonons). In regions of high T , excitations in the neighborhood of the minimum of $\varepsilon(p)$ @ $p = p_0$ are most important. Expanding $\varepsilon(p)$ in powers of $p - p_0$ yields:

$$\varepsilon(p) = \Delta + \frac{1}{2m^*}(p - p_0)^2\tag{28}$$

In (27) elementary excitations are called "rotons." All thermodynamic functions can be expressed as a sum of "phonon-parts" & "roton parts."

To find the thermodynamic potential Ω we substitute eqs. (1) & (27) into the following formula:

$$\Omega = VT \int \ln \left[1 - e^{(\mu - \varepsilon(p))/T} \right] \frac{d\mathbf{p}}{(2\pi)^3} \quad (29)$$

Note that the $\#$ of excitations is not fixed, but is itself determined by the equilibrium condition.

Definition 2.2. Equilibrium Condition: The free energy is a minimum with respect to changes in the $\#$ of particles.

$$\left(\frac{\partial F}{\partial N} \right)_{V,T} = \mu = 0 \quad ; \quad F = (\Omega + \mu N) \quad (30)$$

@ $\mu = 0$, Ω coincides with the free energy F . The energy of the rotons, E_{rot} , is always $\gg$ the temperature. The Bose distribution can therefore be replaced with a Boltzmann distribution. Because $T \ll E_{\text{rot}}$, we can focus only on the first order term in the expansion of:

$$\ln(1 - e^{-E_{\text{rot}}/T}) \quad (31)$$

with respect to a small quantity of $e^{-E_{\text{rot}}/T}$. This leads us to the Boltzmann formula:

$$F_{\text{rot}} = -VT \int e^{-E_{\text{rot}}/T} \frac{d\mathbf{p}}{(2\pi)^3} \quad (32)$$

From this, we gather that:

$$F_{\text{phonon}} = \frac{-V \cdot \pi^2 T^4}{90u^3} \quad (33)$$

$$F_{\text{rot}} = \frac{-V \cdot 2m^{*1/2} T^{3/2} p_0^2}{2\pi^{3/2}} e^{-\Delta/T} \quad (34)$$

These allow us to obtain all other thermodynamic functions.

3. SUPERFLUIDITY

The most interesting property of a Bose liquid is that of superfluidity.

Definition 3.1. Superfluidity: The process of moving through capillary tubes without friction.

To understand this, we consider a Bose liquid @ absolute zero, flowing with velocity $\mathbf{v}$ inside a capillary. In a coordinate system fixed with respect to the liquid, the liquid is at rest when the capillary moves with velocity $-\mathbf{v}$. During this motion, the friction that is caused between the liquid & the capillary wall will allow the liquid to acquire some motion (from the wall). Therefore, the liquid acquires a non-zero energy & momentum, which can only arise through elementary excitations. Let the energy of a liquid be:

$$\varepsilon + \mathbf{p} \cdot \mathbf{v} + \frac{1}{2} M v^2 \quad (35)$$

↑ excitation changes the energy by this much

For an excitation to occur, the change in energy must be negative. It always takes a positive ε to create excitations in a fluid. If the fluid is moving, its excitations also carry a velocity, & therefore some quantity of momenta. For the liquid to be slowed down, there needs to be a decrease in the energy. $\therefore \mathbf{p} \cdot \mathbf{v}$ must be a negative quantity:

$$\varepsilon + \mathbf{p} \cdot \mathbf{v} < 0$$

Geometrically, we can envision this as $\mathbf{p}$ & $\mathbf{v}$ travelling in opposite directions. Thus,

$$\begin{aligned}\varepsilon - pv &< 0 \\ \varepsilon &< pv \\ v &> \frac{\varepsilon}{p}\end{aligned}$$

In order for excitation to appear, the following condition must be met:

$$v > \left(\frac{\varepsilon}{p}\right)_{\min} \quad (36)$$

The minimum value of ε/p occurs @:

$$\frac{d\varepsilon}{dp} = \frac{\varepsilon}{p} \quad \} \text{ velocity of excitation energy} \quad (37)$$

We gather that superfluidity can only occur when the velocity is less than the elementary excitation @ a particular energy. In the case of He^4 , the minimum velocity is not in the low-temperature region, dominated by phonons. It is in the high-temperature region dominated by rotons.

Recall the energy in high-temperature regions @ $p = p_0$ by eq. (28):

$$\varepsilon(p) = \Delta + \frac{1}{2m^*}(p - p_0)^2$$

Recall we denoted the term $\frac{d\varepsilon}{dp}$ in (37) as the velocity. In order to satisfy the inequality (36), we can start by taking the derivative of (28) with respect to p .

$$\begin{aligned}\frac{d\varepsilon}{dp} &= 0 + \frac{1}{2m^*} \cdot 2(p - p_0)(1) \\ &= \frac{p - p_0}{m^*}\end{aligned}$$

By (36),

$$\begin{aligned}\frac{p - p_0}{m^*} &= \frac{\varepsilon}{p} \\ \frac{p - p_0}{m^*} &= \frac{\Delta + \frac{1}{2m^*}(p - p_0)^2}{p}\end{aligned}$$

solving for p yields:

$$\begin{aligned}p - p_0 &= \frac{\Delta + \frac{1}{2m^*}(p - p_0)^2}{p} \cdot m^* \\ p(p - p_0) &= m^* \left(\Delta + \frac{1}{2m^*}(p - p_0)^2 \right)\end{aligned}$$

$$p^2 - pp_0 = \Delta m^* + \frac{1}{2}(p - p_0)^2$$

expanding $(p - p_0)^2$:

$$\begin{aligned} (p - p_0)^2 &= (p - p_0)(p - p_0) \\ &= p^2 - 2pp_0 + p_0^2 \end{aligned}$$

substitution into the above expression yields: substitution into the above expression yields:

$$\begin{aligned} p^2 - pp_0 &= \Delta m^* + \frac{1}{2} [p^2 - 2pp_0 + p_0^2] \\ p^2 - pp_0 &= \Delta m^* + \frac{p^2}{2} - pp_0 + \frac{p_0^2}{2} \\ p^2 - \frac{p^2}{2} &= \Delta m^* + \frac{p_0^2}{2} \\ \frac{2p^2}{2} - \frac{p^2}{2} &= \Delta m^* + \frac{p_0^2}{2} \\ \frac{p^2}{2} &= \Delta m^* + \frac{p_0^2}{2} \\ p^2 &= 2\Delta m^* + p_0^2 \end{aligned}$$

Taking the square root of both sides yields:

$$p = \sqrt{2\Delta m^* + p_0^2}$$

again, by (36), we found that $v = \frac{p-p_0}{m^*}$

substituting our result for p into the above equation for v yields:

$$\begin{aligned} v &= \frac{p - p_0}{m^*} \\ &= \left(\sqrt{2\Delta m^* + p_0^2} - p_0 \right) \cdot \frac{1}{m^*} \end{aligned} \tag{38}$$

Recall that $p = \sqrt{p_0^2 + 2m^*\Delta}$. Factoring out p_0^2 yields:

$$p = p_0 \sqrt{1 + \frac{2m^*\Delta}{p_0^2}}$$

Since $p_0^2 \gg 2m^*\Delta$, we can let $x = \frac{2m^*\Delta}{p_0^2}$.

Recall the Taylor expansion such that:

$$\sqrt{1+x} \approx 1 + \frac{x}{2}$$

$\therefore$

$$p \approx p_0 \left(1 + \frac{2m^*\Delta}{2p_0^2} \right)$$

substitution of this result back into $v = \frac{p-p_0}{m^*}$ yields:

$$v = \frac{p_0 + \frac{m^*\Delta}{p_0} - p_0}{m^*}$$

$$v = \frac{\Delta}{p_0} \tag{39}$$

3.1. Examining the effect on the liquid's motion of excitations already present in the liquid

We assume that a “gas of elementary excitations” move within the liquid with some macroscopic velocity $\mathbf{v}$. The momentum of gas per unit volume is given by:

$$\mathbf{P} = \int \mathbf{p} n(\varepsilon - \mathbf{p} \cdot \mathbf{v}) \frac{d\mathbf{p}}{(2\pi)^3} \quad (40)$$

Recall the Taylor expansion such that:

$$f(x + \Delta x) = f(x) + \Delta x f'(x) + \dots$$

Let

$$\begin{aligned} f &= n \\ x &= \varepsilon \\ \Delta x &= -\mathbf{p} \cdot \mathbf{v} \end{aligned}$$

$$\therefore n(\varepsilon - \mathbf{p} \cdot \mathbf{v}) = \underbrace{n(\varepsilon)}_{\substack{0; \text{ this is due to} \\ \text{the isotropic} \\ \text{integral}}} - (\mathbf{p} \cdot \mathbf{v}) \frac{\partial n}{\partial \varepsilon}$$

$\therefore \mathbf{P}$ becomes:

$$\begin{aligned} \mathbf{P} &= - \int \mathbf{p} (\mathbf{p} \cdot \mathbf{v}) \frac{\partial n}{\partial \varepsilon} \frac{d\mathbf{p}}{(2\pi)^3} \\ &= - \int p^2 \mathbf{v} \frac{\partial n}{\partial \varepsilon} \frac{d\mathbf{p}}{(2\pi)^3} \cdot \underbrace{\frac{1}{3}}_{\substack{3 \text{ directions} \\ \text{present}}} \\ &= - \frac{\mathbf{v}}{3} \int p^2 \frac{\partial n}{\partial \varepsilon} \frac{d\mathbf{p}}{(2\pi)^3} \end{aligned} \quad (41)$$

The main conclusion gathered is that the excitation of a gas relative to the liquid is accompanied by the transport of mass.

To calculate the “normal” density, which is the coefficient of proportionality between the momentum & velocity of a unit volume of the moving gas of excitations, recall eq. (6):

$$\varepsilon = up \rightarrow p = \varepsilon/u$$

Substitution of this result into (40) yields:

$$-\frac{\mathbf{v}}{3} \int \left(\frac{\varepsilon}{u}\right)^2 \frac{\partial n}{\partial \varepsilon} \frac{d\mathbf{p}}{(2\pi)^3} \quad //$$

Also recall (27):

$$\varepsilon(p) = \Delta + \frac{1}{2m^*} (p - p_0)^2$$

substitution into the Boltzmann distribution yields:

$$n \approx e^{-\frac{1}{k_B T}(\Delta + \frac{1}{2m^*}(p-p_0)^2)} \rightarrow e^{-\Delta/k_B T} \cdot e^{-\frac{1}{k_B T}(\frac{1}{2m^*}(p-p_0)^2)}$$

$$n \approx e^{-\frac{1}{k_B T}(\varepsilon(p))} \quad //$$

Finding $\frac{\partial n}{\partial \varepsilon}$:

$$\frac{-1}{k_B T} e^{-\frac{1}{k_B T}(\varepsilon)} \quad //$$

Supplying this result back to the integrand yields:

$$= -\frac{\mathbf{v}}{3} \int \left(\frac{\varepsilon}{u}\right)^2 \left(-\frac{1}{k_B T} e^{-\frac{\varepsilon}{k_B T}}\right) \frac{d\mathbf{p}}{(2\pi)^3}$$

From $\varepsilon = up$, we can modify the integrand:

$$= -\frac{\mathbf{v}}{3} \int \left(\frac{up}{u}\right)^2 \left(-\frac{1}{k_B T} e^{-(p-p_0)^2/2m^*k_B T} e^{-\Delta/k_B T}\right) \frac{d\mathbf{p}}{(2\pi)^3}$$

$$= -\frac{\mathbf{v}}{3} \left(-\frac{1}{k_B T}\right) \frac{1}{(2\pi)^3} e^{-\Delta/k_B T} \int p^2 e^{-(p-p_0)^2/2m^*k_B T} d\mathbf{p} \quad //$$

Breaking the integrand into a product (keep α in mind)

$$\int p^4 dp e^{-(p-p_0)^2/2m^*k_B T} \underbrace{\int_0^\pi \sin \theta d\theta}_2 \underbrace{\int_0^{2\pi} d\phi}_{2\pi}$$

$$\alpha = -\frac{\mathbf{v}}{3} \left(-\frac{1}{k_B T}\right) \left(\frac{1}{(2\pi)^3}\right) \cdot 4\pi$$

$$\int p^4 e^{-(p-p_0)^2/2m^*k_B T} dp$$

For the roton, recall $p = p_0$. $\therefore$

$$\int_{-\infty}^{\infty} p_0^4 e^{-(p-p_0)^2/2m^*k_B T} dp$$

Recall the Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-\lambda x^2} dx = \sqrt{\frac{\pi}{\lambda}}$$

From the integrand, let $x = (p - p_0)$; $\lambda = \frac{1}{2m^*k_B T}$

$$\therefore p_0^4 \int_{-\infty}^{\infty} e^{-(p-p_0)^2/2m^*k_B T} dp$$

$$= \frac{1}{\sqrt{\frac{1}{2m^*k_B T}}} = \sqrt{\pi \cdot 2m^*k_B T} = \sqrt{2m^*k_B T \pi}$$

Multiplying all the results with α , we obtain:

$$-\frac{\mathbf{v}}{3} \cdot \left(-\frac{1}{k_B T}\right) \frac{1}{(2\pi)^3} \cdot 4\pi \cdot p_0^4 \cdot \sqrt{2m^*k_B T \pi} \cdot e^{-\Delta/k_B T}$$

$$\begin{aligned}
& \frac{\mathbf{v}}{3} \cdot \frac{1}{k_B T} \frac{1}{(2\pi)^3} \cdot 4\pi \cdot p_0^4 \sqrt{2\pi \cdot k_B T m^*} \cdot e^{-\Delta/k_B T} \\
& \frac{\mathbf{v}}{3} \cdot k_B T^{-1} (2\pi)^{-3} \cdot 4\pi \cdot p_0^4 \cdot 2^{1/2} \cdot k_B T^{1/2} \cdot m^{*1/2} \cdot e^{-\Delta/k_B T} \cdot \pi^{1/2} \\
& \frac{\mathbf{v}}{3} e^{-\Delta/k_B T} p_0^4 m^{*1/2} \pi^{3/2} k_B T^{-1/2} \cdot 2^{5/2} \cdot (2\pi)^{-3} \\
& \frac{\mathbf{v}}{3} e^{-\Delta/k_B T} p_0^4 m^{*1/2} (2\pi)^{-3/2} k_B T^{-1/2} \\
& \frac{p_0^4 m^{*1/2} e^{-\Delta/k_B T} \mathbf{v}}{3(2\pi)^{-3/2} (k_B T)^{1/2}} \quad // \quad \text{if we neglect } k_B \text{ (for whatever unspecified reason), we obtain the same result.}
\end{aligned}$$

$$\rho_{\text{rot}} = \frac{p_0^4 m^{*1/2} e^{-\Delta/T}}{3(2\pi)^{-3/2} (T)^{1/2}} \quad (42)$$

For the normal density of the phonon, we begin with (40)

$$\mathbf{P} = \int \mathbf{p} n(\varepsilon - \mathbf{p} \cdot \mathbf{v}) \frac{d\mathbf{p}}{(2\pi)^3}$$

for small velocities, we use eq. (41):

$$-\frac{\mathbf{v}}{3} \int p^2 \frac{\partial n}{\partial \varepsilon} \frac{d\mathbf{p}}{(2\pi)^3}$$

Because phonons are bosons, we need a different distribution. The one used in the prior case was “roton-specific.” For the phonon, let:

$$n = \frac{1}{e^{\varepsilon/k_B T} - 1}$$

$$\frac{\partial n}{\partial \varepsilon} = -\frac{1}{k_B T} \left(\frac{e^{\varepsilon/k_B T}}{(e^{\varepsilon/k_B T} - 1)^2} \right)$$

in this regime we only need to apply $\varepsilon = up$ (“phonon-specific”)

$\therefore$ (41) becomes:

$$\begin{aligned}
& -\frac{\mathbf{v}}{3} \int p^2 \left(-\frac{1}{k_B T} \frac{e^{\varepsilon/k_B T}}{(e^{\varepsilon/k_B T} - 1)^2} \right) \frac{d\mathbf{p}}{(2\pi)^3} \\
& \frac{\mathbf{v}}{3} \cdot \frac{1}{k_B T} \cdot \frac{1}{(2\pi)^3} \int p^2 \frac{e^{\varepsilon/k_B T}}{(e^{\varepsilon/k_B T} - 1)^2} \cdot p^2 \sin \theta \, dp \, d\theta \, d\phi \\
& \frac{\mathbf{v}}{3} \cdot \frac{1}{k_B T} \cdot \frac{1}{(2\pi)^3} \int p^4 \frac{e^{\varepsilon/k_B T}}{(e^{\varepsilon/k_B T} - 1)^2} \, dp \underbrace{\int_0^\pi \sin \theta \, d\theta \int_0^{2\pi} d\phi}_{4\pi} \\
& \frac{\mathbf{v}}{3} \cdot \frac{1}{k_B T} \cdot \frac{1}{(2\pi)^3} \cdot 4\pi \int_{-\infty}^{+\infty} \frac{p^4 e^{\varepsilon/k_B T}}{(e^{\varepsilon/k_B T} - 1)^2} \, dp
\end{aligned}$$

In order to evaluate the integrand, let $x = \frac{\varepsilon}{k_B T} = \frac{up}{k_B T}$.

Rearranging for p yields:

$$x = \frac{up}{k_B T} \rightarrow p = \left(\frac{k_B T}{u} \right) x$$

This implies that $dp = \left(\frac{k_B T}{u}\right) dx$.

Substitution of these variables into the integrand yields:

$$\begin{aligned} & \frac{\mathbf{v}}{3} \cdot \frac{1}{k_B T} \cdot \frac{1}{(2\pi)^3} \cdot 4\pi \int_{-\infty}^{+\infty} \left(\frac{k_B T}{u}\right)^4 x^4 \frac{e^x}{(e^x - 1)^2} \left(\frac{k_B T}{u}\right) dx \\ & \frac{\mathbf{v}}{3} \cdot \frac{1}{k_B T} \cdot \frac{1}{(2\pi)^3} \cdot 4\pi \left(\frac{k_B T}{u}\right)^5 \int_{-\infty}^{+\infty} \frac{x^4 e^x}{(e^x - 1)^2} dx \end{aligned}$$

Recall the formula for integration by parts:

$$\int u dv = uv - \int v du$$

Note: we will disregard the pre-factor for now. Let:

$$\begin{aligned} u &= x^4 \quad ; \quad du = 4x^3 \\ dv &= \frac{e^x}{(e^x - 1)^2} dx \quad ; \quad v = \frac{-1}{(e^x - 1)} \\ \int u dv &= \underbrace{\frac{-x^4}{(e^x - 1)}}_0 \Big|_0^\infty - \int -\frac{1}{e^x - 1} (4x^3) dx \\ &= 4 \int \frac{x^3}{e^x - 1} dx \end{aligned}$$

Recall the gamma-zeta identity such that:

$$\int_0^\infty \frac{x^{n-1}}{e^x - 1} dx = \Gamma(n)\zeta(n)$$

From the numerator, we let $n = 4$

$$\begin{aligned} \therefore 4 \int \frac{x^3}{e^x - 1} dx &= 4\Gamma(4)\zeta(4) \\ &= 4(3!) \left(\frac{\pi^4}{90}\right) = \frac{4\pi^4}{15} \end{aligned}$$

Multiplication of this result by the pre-factor yields:

$$\begin{aligned} &= \frac{\mathbf{v}}{3} \cdot \frac{1}{k_B T} \cdot \frac{1}{(2\pi)^3} \cdot 4\pi \left(\frac{k_B T}{u}\right)^5 \left(\frac{4\pi^4}{15}\right) \\ &= \frac{\mathbf{v}}{3} \cdot \frac{1}{8\pi^3} \cdot 4\pi \cdot \frac{1}{k_B T} \cdot \frac{(k_B T)^5}{u^5} \cdot \frac{4\pi^4}{15} \\ &= \mathbf{v} \cdot \frac{(k_B T)^4}{u^5} \cdot \frac{16\pi^5}{8\pi^3} \cdot \frac{1}{45} \\ &= \frac{(k_B T)^4}{u^5} \cdot 2\pi^2 \cdot \frac{1}{45} \end{aligned}$$

Once again neglecting the Boltzmann constant yields the following for the normal density of the phonon:

$$\rho_{\text{phonon}} = \frac{2\pi^2 T^4}{45u^5} \quad (43)$$

Beyond (42) & (43), the remaining part of the density is denoted as:

$$\rho = \rho_n + \rho_s \quad (44)$$

An important feature of the velocity of a superfluid is that its rotation is irrotational. That is:

$$\nabla \times \mathbf{v}_s = 0 \quad (45)$$

The motion of the superfluid imposes certain conditions on the excitations. In the rest system, we have:

$$\varepsilon' = \varepsilon(\mathbf{p}) + \mathbf{p} \cdot \mathbf{v}_s \quad (46)$$

The transport equation for the excitations is:

$$\frac{\partial n}{\partial t} + \nabla_{\mathbf{r}} n \cdot \nabla_{\mathbf{p}} \varepsilon' - \nabla_{\mathbf{p}} n \cdot \nabla_{\mathbf{r}} \varepsilon' = \underbrace{I(n)}_{\text{collision integral}}$$

The Bose liquid contains two types of motions with different velocities. In turn, two different kinds of oscillations can occur. The first kind is when velocity propagation is equal to u . Sound waves of the first kind travel as a whole, $\therefore$ the normal & superfluid parts do not separate.

As for the second kind, velocity is:

$$u_2 = \sqrt{\frac{\rho_s T S^2}{\rho_n \rho C}} \quad (47)$$

where C & S are heat capacity & entropy respectively. The oscillations of the normal & superfluid part have opposite phases. The total vector flow of the liquid is:

$$\mathbf{j} = \rho_n \mathbf{v}_n + \rho_s \mathbf{v}_s = 0 \quad (48)$$

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